Low temperature in Medicine
Cryogenics, Cryosurgery, and Radiotherapy

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Objectives

Students will understand the principles of cryogenics and cryosurgery, including the methods of cooling, the benefits and applications of cryogenic technology in medicine, and the significance of blood preservation and treatment of various cancers through cryosurgery.

Recommended text book:

1. Medical Physics for Medical students (Taki AM Almusa) 2024
2. Medical physics (John R. Cameron) 1978 ,1993,1999,2003 and 2010

3. Introduction of Health Physics (Herman Cember) 2009

4. Introduction to physics in modern medicine (Suzanne A. kane) 2010

5. Electronics in medicine and biomedical instrumentation (Nandini K. Jog) 2006

6. Radiation protection and dosimetry (Michel G. Stabin) 2012

7. Dr.R.N.Roy .Atext book of Biophysics 1st edition,2005

8- Physics of radiotherapy Khan 2012

9- Advance in Medical Physics, Devon J. Godfrey, Volume 6 2016

Cryogenics:-

It is the science and technology of producing and using very low temperature

Cryogenics utilizes temperatures below (-150^0C to $-273\ ^0$C) to manipulate materials and biological tissues, with critical medical applications .

Medical Applications

Cryopreservation: Biological specimens (e.g., sperm, embryos, eggs, stem cells) are preserved for long-term storage in liquid nitrogen (-196 ^{0}C) to prevent cellular degradation.

Cryosurgery/Cryotherapy: Extremely low temperatures are applied to kill diseased or abnormal tissues, such as skin cancer, skin tags, and warts.

Organ Transplantation: Cryogenic engineering assists in preserving organs for longer periods by slowing down metabolism and cellular degeneration.

Medical Imaging (MRI): Superconducting magnets in MRI machines require liquid helium to maintain temperatures near (-269 ^{0}C) to function.

Cryogenic Techniques

Liquid Nitrogen (LN_2): Most commonly used for rapid freezing (-196 ^{0}C) in dermatology and specimen storage.

Cryospray/Cryoprobes: Tools that deliver freezing agents directly to tissue, destroying it through ice formation and cellular damage.

Whole-body Cryotherapy: Exposure to extreme cold (-110 ^{0}C to -140 ^{0}C) to reduce inflammation and manage pain.

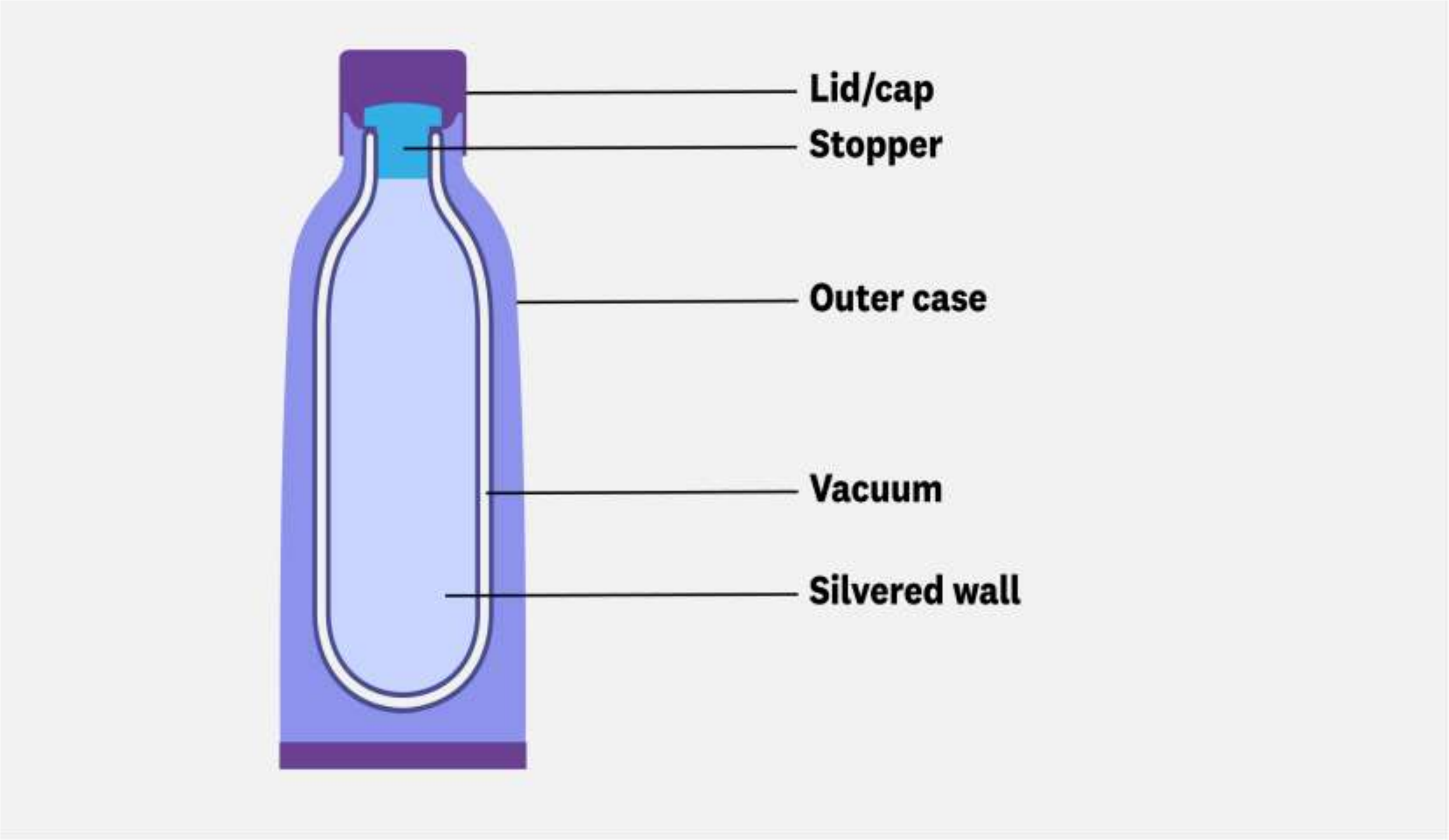
These cold liquids have many medical and biological advantages.
The storage of liquefied gases is rather difficult because it can take heat rapidly from the environment due to the large difference in temperature.

Liquefied gas storage requires specialized, robust containment to maintain liquids under high pressure or cryogenic temperatures, using double-walled tanks, Dewar vessels

An excellent example of the design used to reduce the heat transfer to the surroundings is a **thermos flask** (**Dewar container**). To keep the liquid hot or to keep the liquid cold, we use **thermos flasks**, also known as **Dewar flasks**. The vacuum flask consists of two containers, one placed within the other and joined at the neck.

The three essential features that help thermal flasks reduce the rate of thermal energy transfer (heat loss or gain) are:-

- The double-wall vacuum insulation
- The reflective inner surface
- The tight-sealing stopper.



Dewar flasks

Blood Types

O+ is found in 37 percent of the population.

A+ is found in 34 percent of the population.

(A-, B+, B-) less than 10 percent of the population

O- is found in six percent of the population.

AB+ less than four percent

A- less than 1% of the population

For conventional blood storage it can be stored with anticoagulant at 4°C. About 1% of the red blood cells hemolysis (break) each day so the blood will not be suitable for use after 21 days.

For rare blood types should be stored for longer periods of time (for up to 30 years and thawed for transfusion when it's needed).

<u>There are two ways to freeze the blood to (-196°C): -</u>

(1) The blood sand method in this method the blood sprayed on the surface of liquid nitrogen and then it will be frozen in small droplets in very short time forming sand like particles, then stored at liquid nitrogen temperature.

(2) The blood is kept in a thin wall, with large surface area metal container and the spacing between the walls of the container is small to maintain a small thickness of blood inside the container . The container with the blood is immersed into the liquid nitrogen making very rapid cooling.

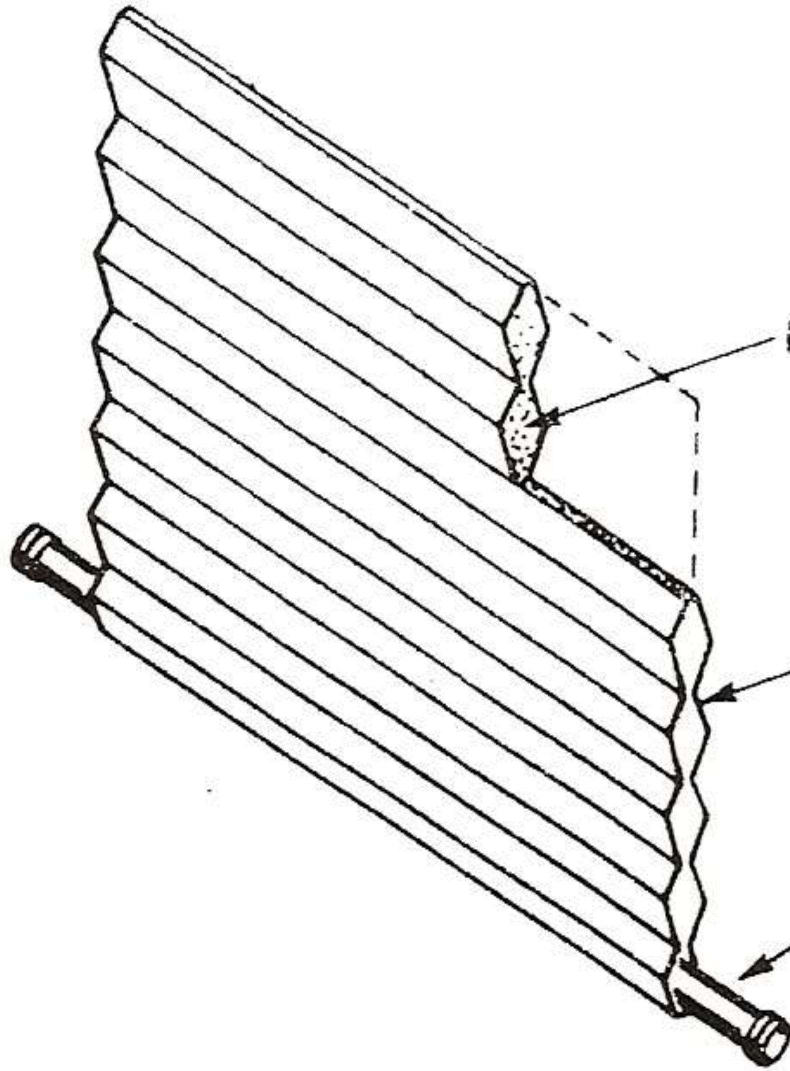


Figure 4.13. Container for whole-blood freezing at cryogenic temperatu
***Cryogenic Systems* by R. Barron. Copyright © 1966, McGraw-Hill, New Y**
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Cryosurgery:

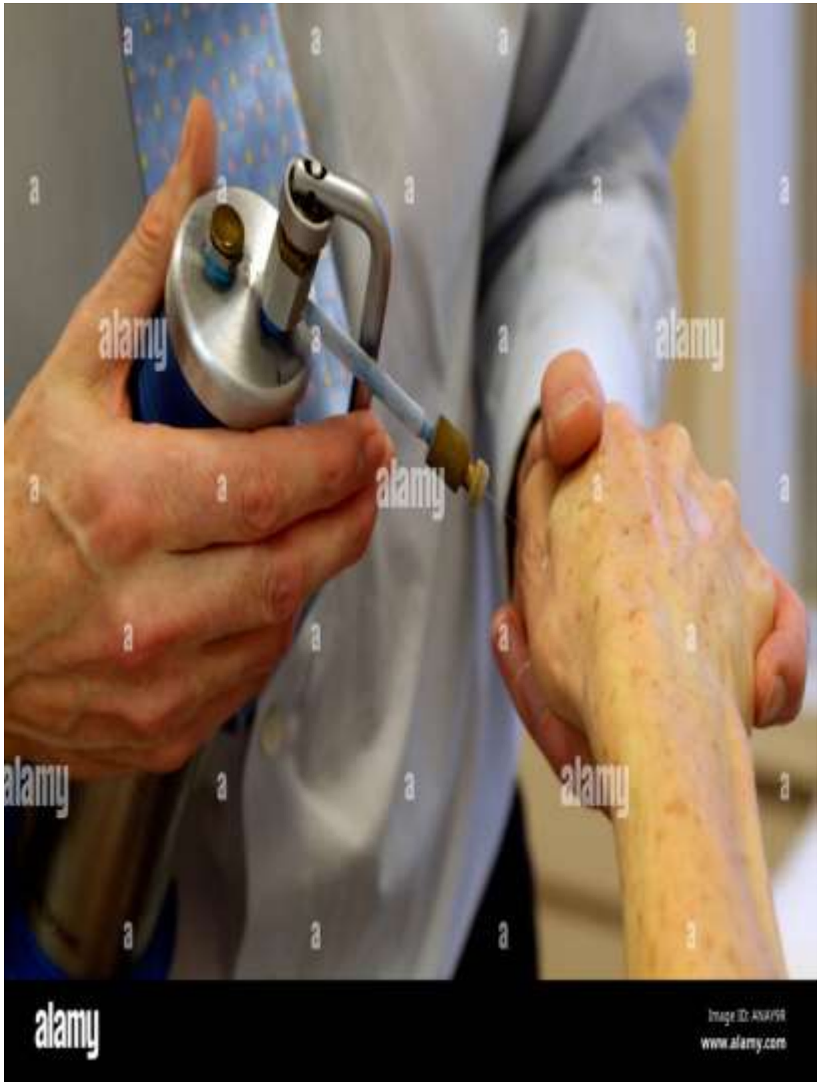
The branch of surgery applying very low temperatures (down to-196° C) to destroy malignant tissue, cancer cells. It can only be used in small area, and cannot be used to treat cancer that has spread beyond the gland itself, or to distant areas which the cryoprobe could not reach easily.

This type of surgery destroys cancer cells by freezing them . It has several and advantages: -

1. **Cause a little bleeding.**
2. **The volume of the tissue destroyed can be controlled.**
3. **Little pain because low temperature desensitize nerves.**
4. **Very short recovery.**

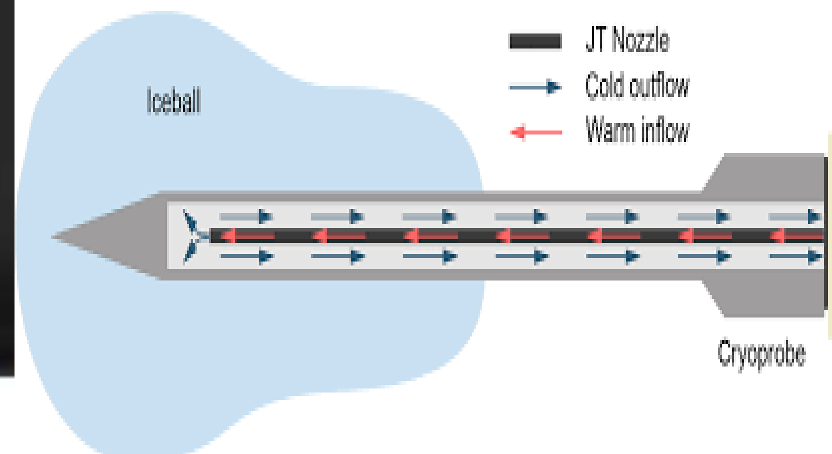
Cryosurgery may be recommended for many types of cancer:

- Parkinson disease (Shaking palsy).
- Skin cancer
- Liver cancer
- Cervical cancer
- Prostate cancer that is only in the prostate
- Bone cancer
- Lung cancer, so forth



The gas in Cryosurgery is circulated through a hollow instrument called a cryoprobe, which is placed in contact with the tumor.

Cryotherapy apparatus. Liquid nitrogen unit with cryoprobe attachment



The cryotube is then in direct contact to the tumor, the procedure will utilize an ultrasound or a MRI to guide the cryoprobe and monitor the freezing process to minimize the damage done to the healthy tissue.

Cryosurgery is used to treat external growths on the skin (**skin tags, and warts**), and it is not only used outside the body, but can be used for tumors growing inside the body

Parkinson's disease (PD)

One of the first uses of cryosurgery is in the treatment of Parkinson disease (Shaking palsey).

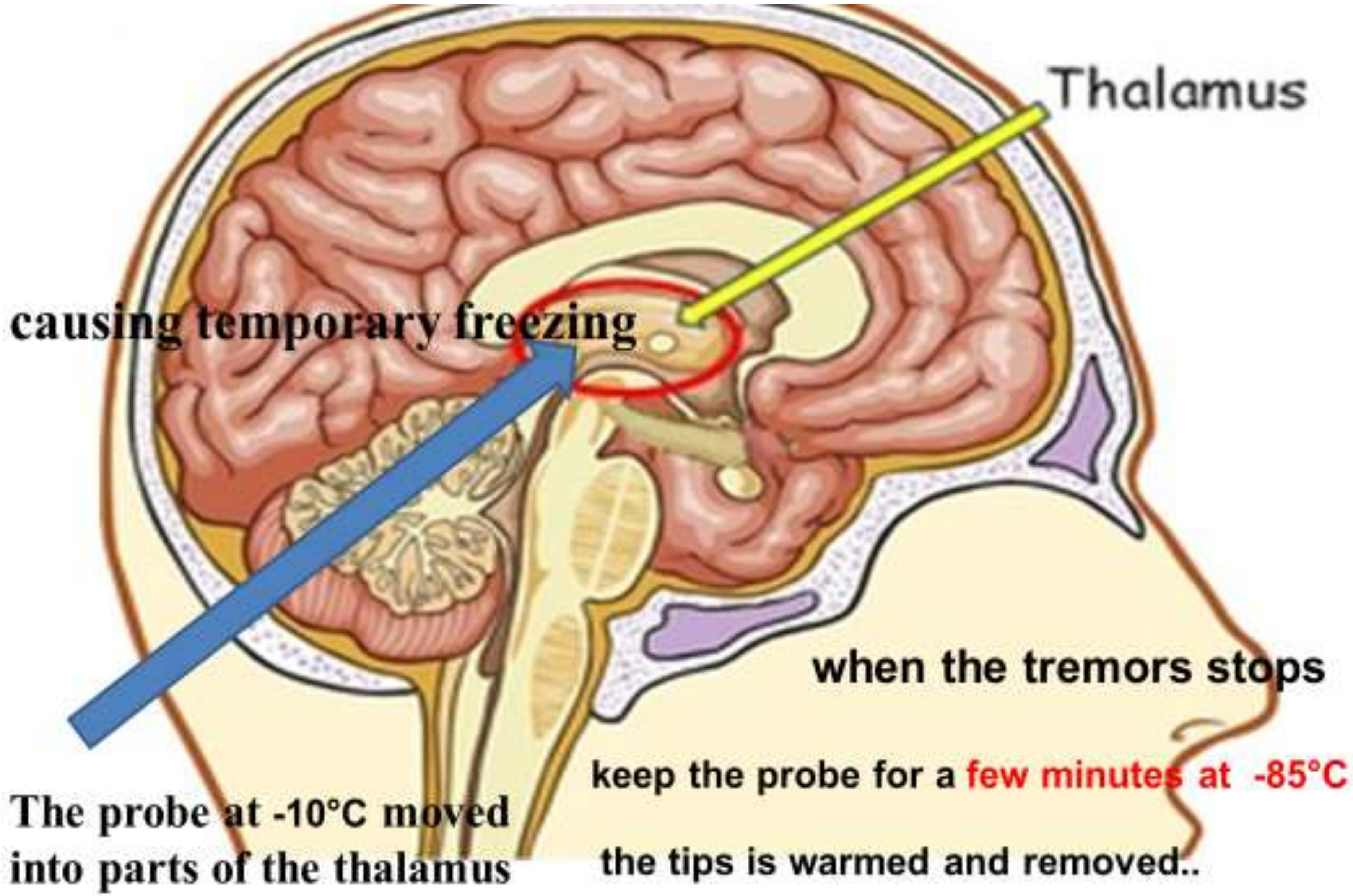
-This disease cause uncontrolled tremors in the arms and legs.

-It is possible to stop it by destroying parts of the thalamus of the brain that controls nerve impulse to the other part of the nervous system.

- The probe at -10°C moved into the appropriate parts of the thalamus causing temporary freezing.

-The frozen area can recover if the probe tip is removed in less than 30 sec.

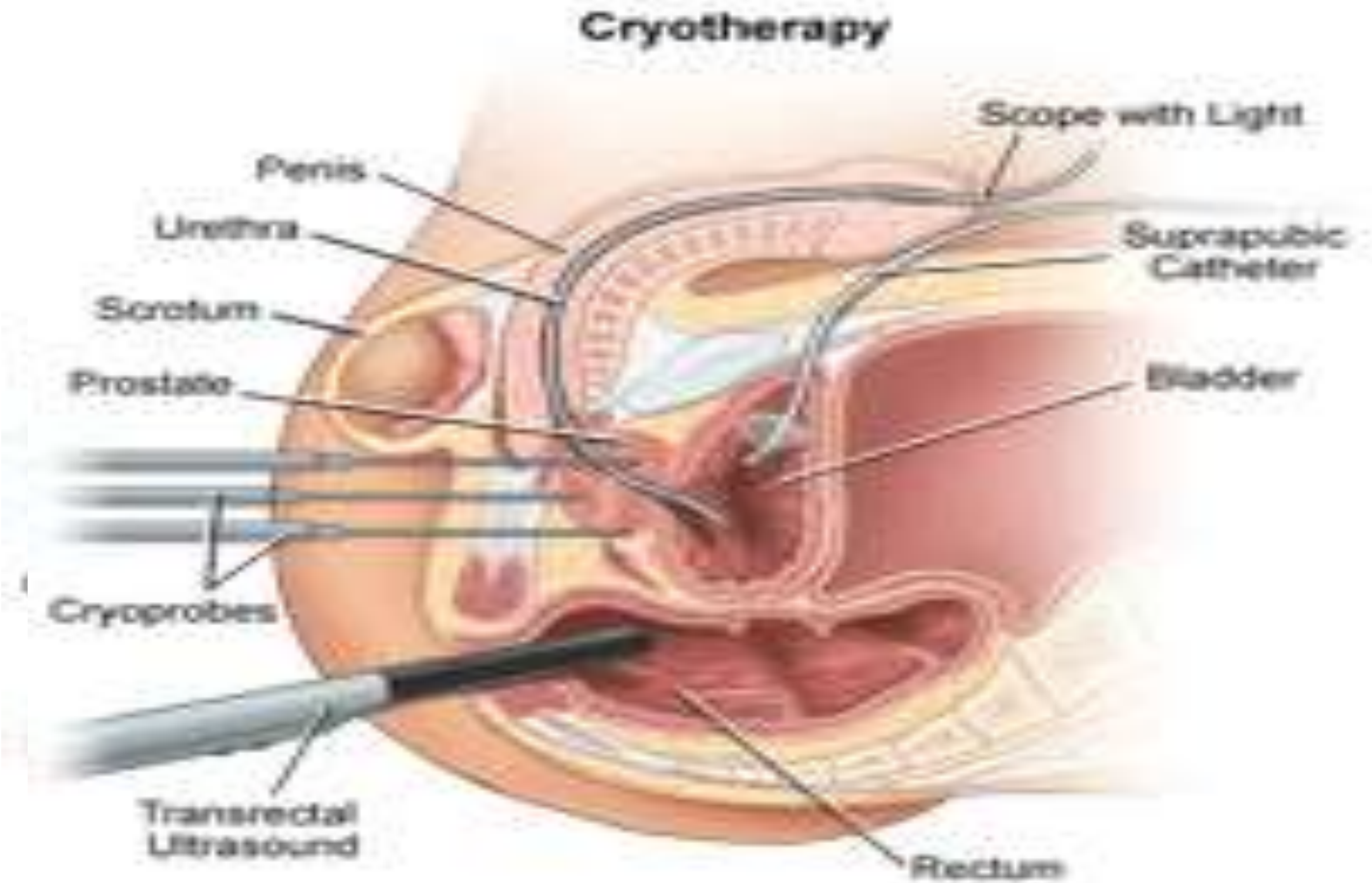
- When the tremors stops, keep the probe for a few minutes at -85 ^{0}C , and then the tip is wormed and removed

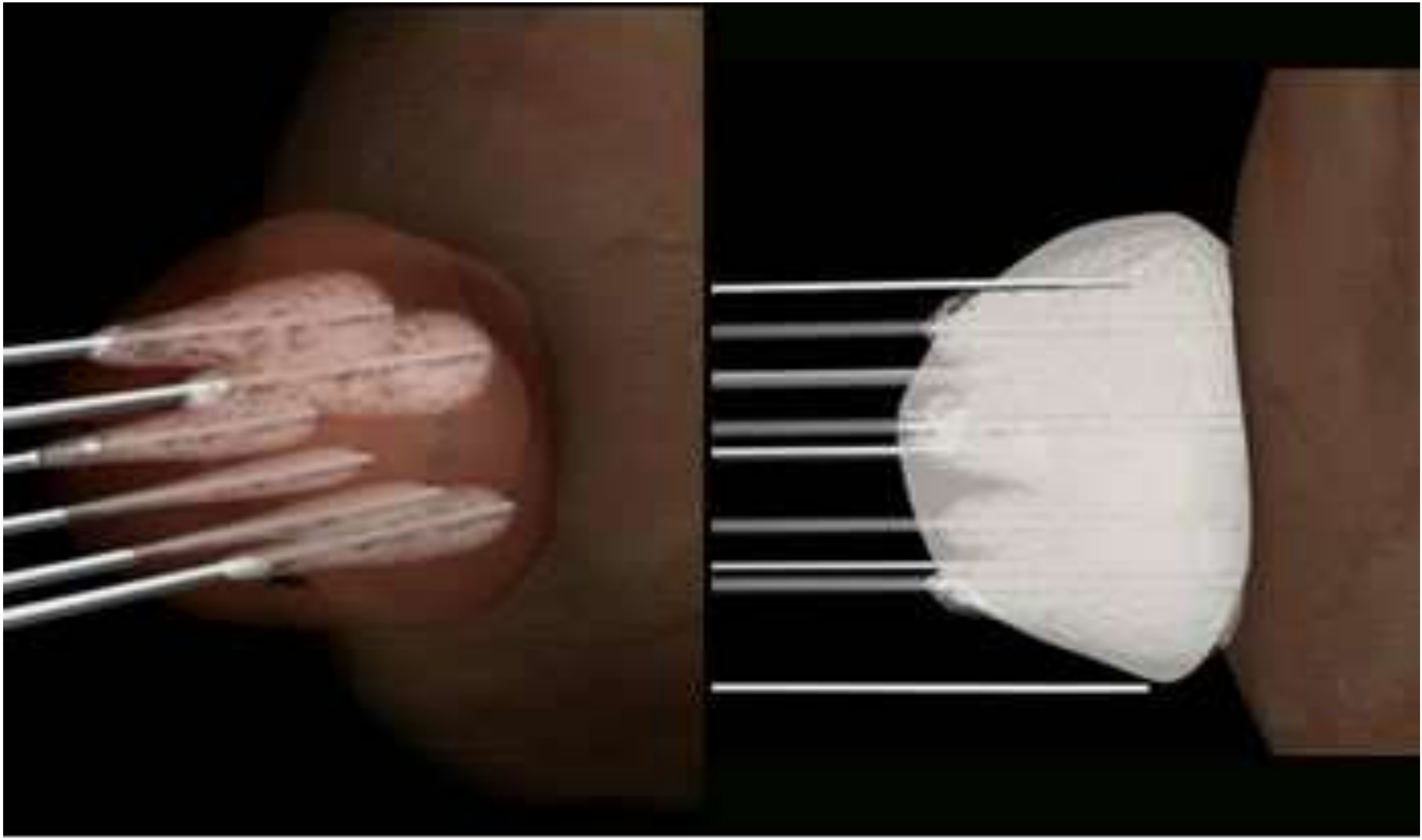


Thalamus
causing temporary freezing
when the tremors stops
keep the probe for a few minutes at -85°C
The probe at -10°C moved
into parts of the thalamus
the tips is warmed and removed..

Prostate cryosurgery

Cryosurgery can be used to treat early-stage prostate cancer, but only when it is confined to the prostate gland

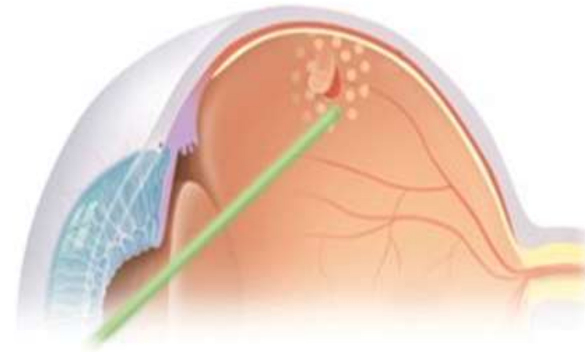




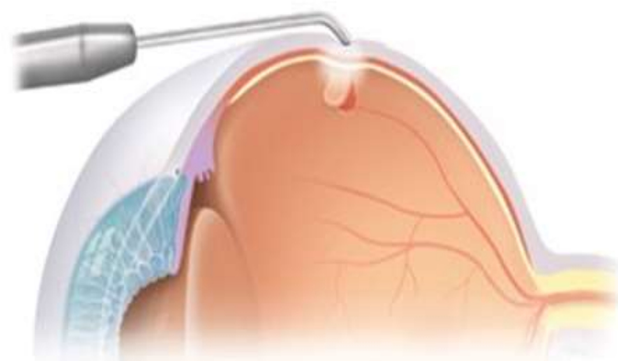
Prostate cryosurgery freezes prostate tissue by creating very cold temperatures on the outside of the cryoprobe causing the prostate tissue to form an ice ball, as the ice balls grow they turn the entire prostate into a frozen mass.

Cryosurgery is used in several types of eye surgery: -

1- In retina detachment

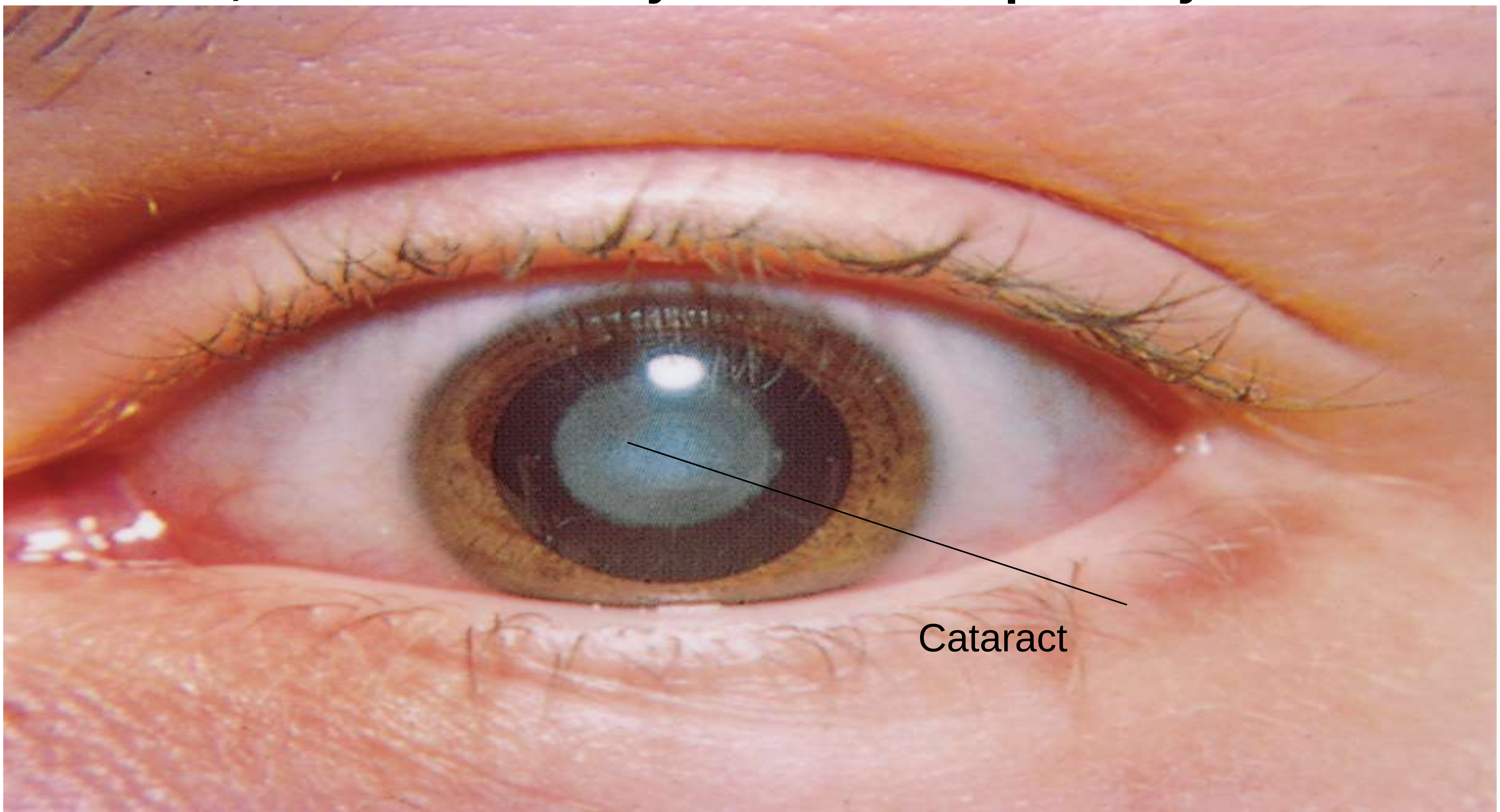


The most common symptom of retinal detachment is a gradual deterioration in vision. It's often described as a shadow spreading across the vision of one eye. It may only affect parts of your vision at first.



In retina detachment a cooled tip is applied to the outside of the eyeball in the vicinity of the detachment a reaction occurs that acts to weld the retina to the wall of the eyeball.

2- The cryosurgical extract of the lens (during cataract)
In this procedure the cold probe is touched to the front of the lens. The probe sticks to the lens making the lens easy to remove. cataract an abnormal progressive condition of the lens, characterized by loss of transparency



Radiation Therapy

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Radiotherapy, or radiation therapy, uses high-energy radiation (like X-rays, gamma rays, protons) to kill cancer cells or slow their growth by damaging their DNA, a common cancer treatment used alone or with surgery/chemo to cure cancer, shrink tumors, or relieve symptoms, with side effects varying by treated area but often including fatigue and skin changes.

Types of radiotherapy

External Beam Radiation: Radiation comes from a machine outside the body (linear accelerator).

Internal Radiation (Brachytherapy): Radioactive material is placed inside the body near the cancer.

Systemic Radiation: A radioactive substance travels through the blood (e.g., radiolabeled antibodies).

Types of External Beam Radiation Therapy

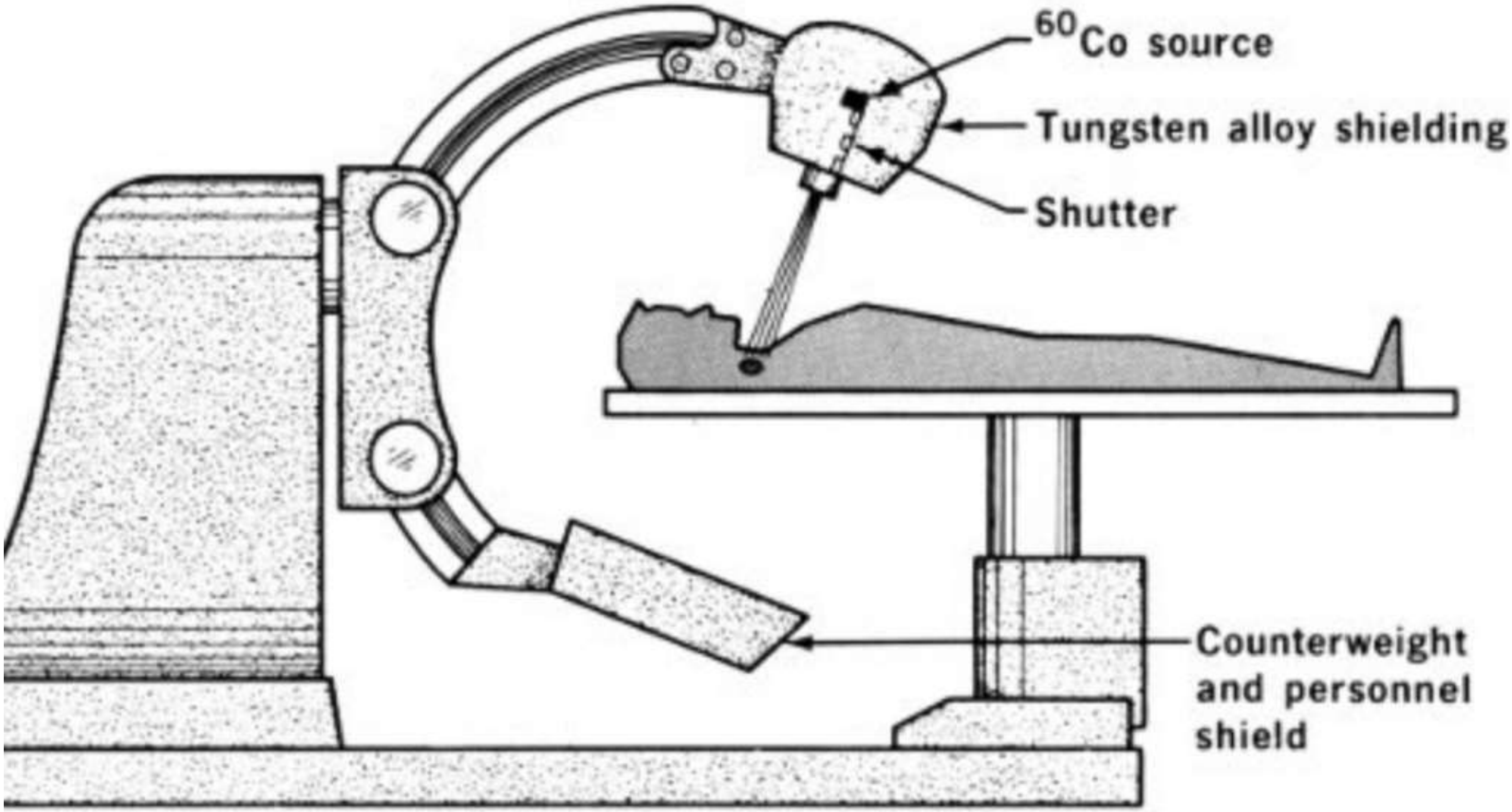
Cobalt therapy is a Gamma Rays emitted from Cobalt 60 machine

Two-dimensional radiation therapy

Three-dimensional conformal radiation therapy (3-D CRT)

Intensity modulated radiation therapy (IMRT)

Volumetric Modulated Arc Therapy VAMAT


^{60}Co source
Tungsten alloy shielding
Shutter
Counterweight and personnel shield

A machine called a linear accelerator, or linac, creates the radiation beam for x-ray or photon radiation therapy.

Radiotherapy involves a three-step process:-
1-Simulation
2- Planning
3- And treatment—designed to deliver precise radiation to tumors while sparing healthy tissue.

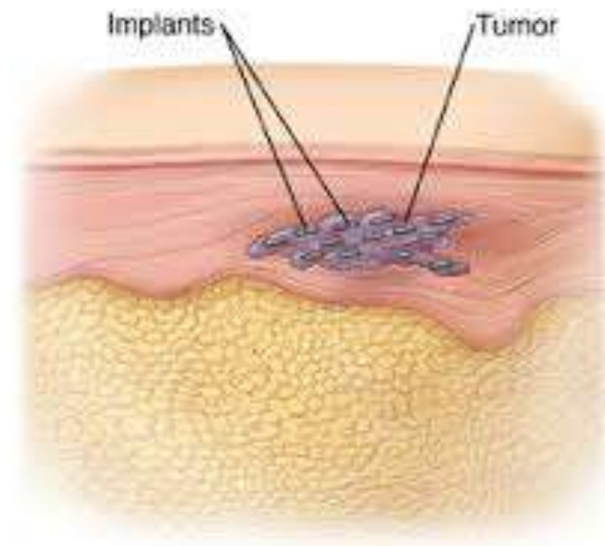
1- Radiotherapy Simulation (The Setup)
Purpose: To determine the exact, reproducible position for treatment and identify the area requiring radiation.

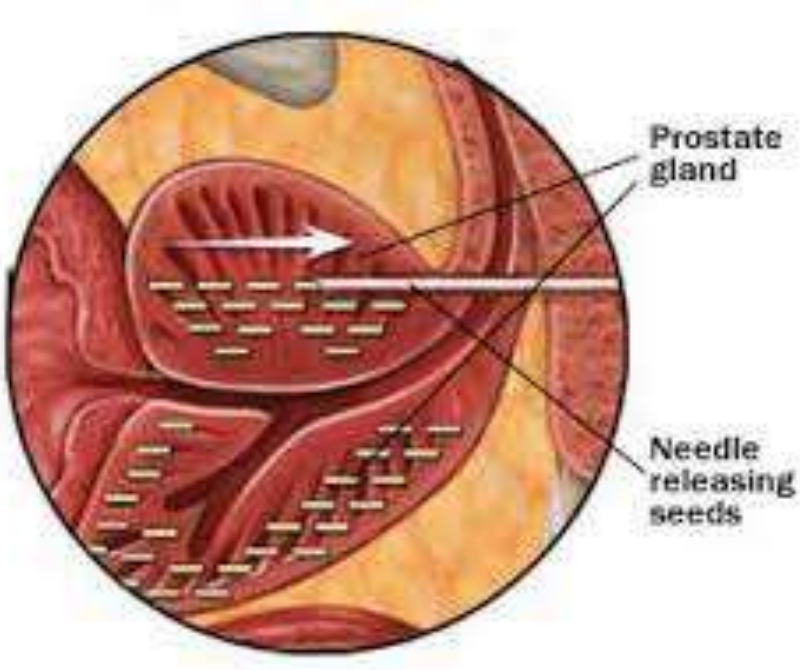
2. Radiotherapy Planning (The Map)
Targeting: Radiation oncologists use simulation CT images to identify the tumor (target volume) and surrounding healthy organs that must be protected.
Dosimetry: Medical dosimetrists and physicists calculate the exact angle, shape, and intensity of radiation beams to ensure maximum dosage to the cancer and minimal exposure to healthy tissue.

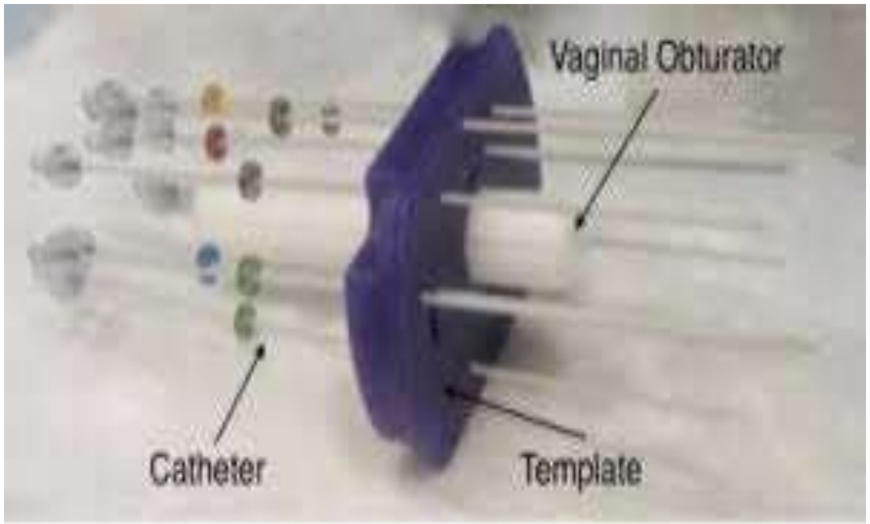
3. Radiotherapy Treatment
Treatment Delivery: A machine, such as a linear accelerator, delivers the radiation. While the session may take 15-30 minutes, the actual radiation delivery is very fast.
Monitoring: The radiation team monitors the patient via cameras and intercom,, adjusting to ensure accuracy, and treatments are painless.

Brachytherapy

Brachytherapy is a type of internal radiation therapy that treats cancer by placing radioactive sources (like seeds, ribbons, or capsules) directly inside or next to a tumor, delivering a high dose of targeted radiation while sparing healthy tissue.

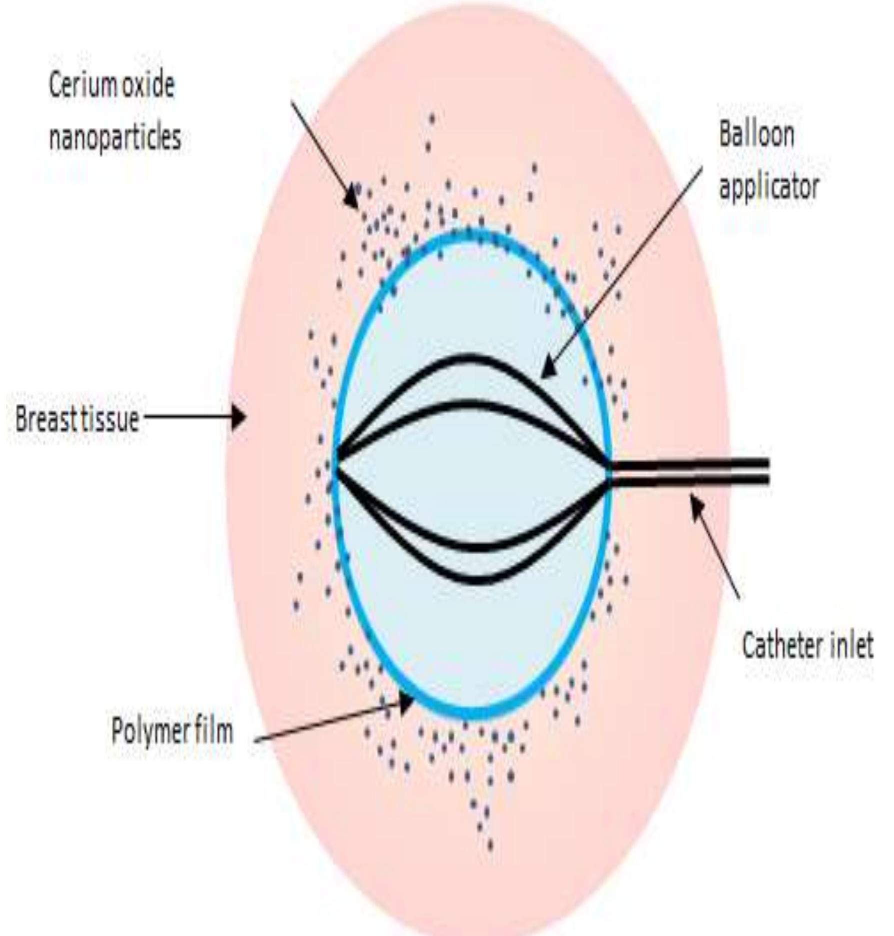






The intracavitary brachytherapy

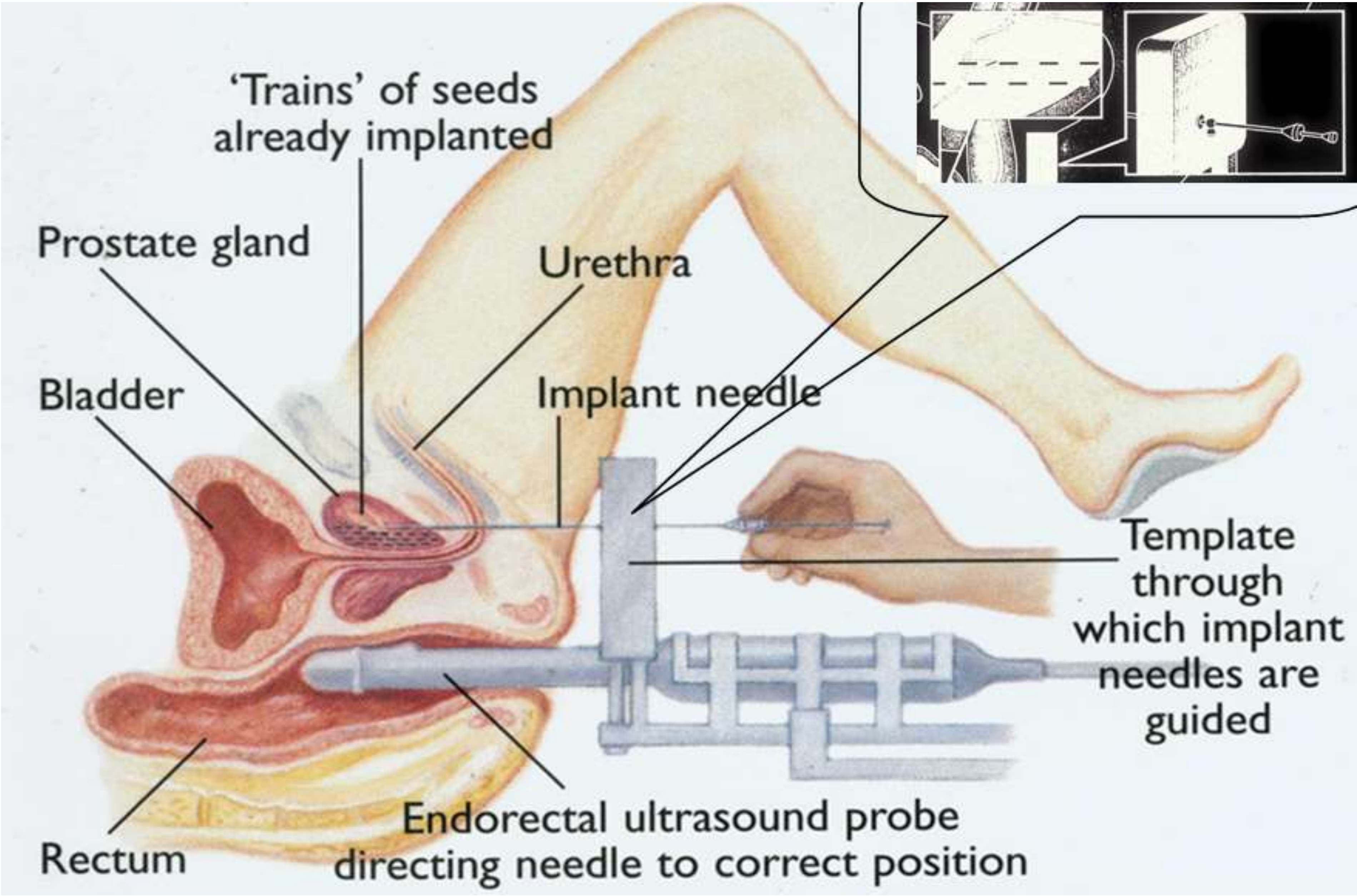
In the intracavitary brachytherapy technique, a balloon is placed in the lumpectomy bed at the time of surgery. The balloon is **inflated** to approximate the walls of the cavity. A source is then placed in the balloon to deliver radiation to the walls of the lumpectomy bed..



Brachytherapy may be delivered via interstitial or intracavitary techniques

The interstitial technique consists of placing hollow needles through the breast and lumpectomy bed in multiple planes.

Internal radiation therapy (Brachytherapy)



Principles of Radiotherapy

1. Delivering of an optimal dose to the tumor

2. Minimal damage of surrounding organs & tissues.

Summary

- Radiation therapy is a well-established modality for the treatment of numerous malignancies
- Radiation oncologists are specialists trained to treat cancer with a variety of forms of radiation
- Treatment delivery is safe, quick and painless

1. What is cryogenics?
2. What is the purpose of a Dewar flask?
3. How long can rare blood types be stored when frozen in liquid nitrogen?
4. What is one advantage of cryosurgery?
5. Which cancer is NOT mentioned as being treated with cryosurgery?
6. What is one method used to freeze blood rapidly?
8. What temperature is used in cryosurgery to destroy malignant tissue?

9. Which type of eye surgery uses cryosurgery?
10. Radiation therapy is used after surgery to:

Thank you